1E 1 MOLAR VOLUME CALCULATIONS

SPECIFICATION REFERENCE
1.8(ii) CP1

LEARNING OBJECTIVES

- Use chemical equations to calculate reacting volumes of gases and vice versa using the concept of molar volume of gases.

MOLAR VOLUME

The work of Avogadro and others led to the idea of **molar volume**, the volume of gas that contains one mole of that gas. The molar volume is approximately the same for all gases, but its value varies with temperature and pressure. The value most often used is for gases at room temperature and pressure (sometimes abbreviated as r.t.p.). Room temperature is 298 K (or 25 °C) and standard pressure varies, but is often quoted as 1.01×10^5 Pa. The value of molar volume is usually quoted as 24 $dm^3\,mol^{-1}$ (equivalent to 24 000 cm^3). Its symbol is V_m.

$V_m = 24\,dm^3\,mol^{-1}$ at r.t.p.

and

$V_m = 24\,000\,cm^3\,mol^{-1}$ at r.t.p.

CALCULATIONS USING MOLAR VOLUME

CALCULATIONS INVOLVING A SINGLE GAS

If you are asked about a single gas, the calculation is straight forward. Assume that in these examples, all volumes are measured at r.t.p.

You need the expression $V_m = 24\,dm^3\,mol^{-1}$ which you might want to consider using in these alternative forms:

$$V_m = 24.0 = \frac{\text{volume in dm}^3}{\text{amount in mol}} \text{ or } V_m = 24\,000 = \frac{\text{volume in cm}^3}{\text{amount in mol}}$$

You will need to rearrange the expression depending on the actual question. Make sure that you use only 24 and dm^3, or only 24 000 and cm^3, in the calculation.

EXAMPLE 1

What is the amount, in moles, of CO in 3.8 dm^3 of carbon monoxide?

$$\text{Answer} = \frac{3.8}{24} = 0.16\,\text{mol}$$

EXAMPLE 2

What is the amount, in moles, of CO_2 in 500 cm^3 of carbon dioxide?

$$\text{Answer} = \frac{500}{24\,000} = 0.021\,\text{mol}$$

EXAMPLE 3

What is the volume of 0.36 mol of hydrogen?

Answer = $24 \times 0.36 = 8.64\,dm^3$

CALCULATIONS INVOLVING GASES AND SOLIDS OR LIQUIDS

You may be given a chemical equation that involves one or more gases and a solid or a liquid. If you use information about the amount (in mol) of a solid or liquid, you can combine two of the calculation methods that we have already used.

The basis of this type of calculation is that for gases you can interconvert between amount and volume. For solids and liquids you can interconvert between amount and mass.

Step 1: Calculate the amount in moles from either the mass or the volume, depending on which one is given.

Step 2: Use the relevant reaction ratio in the equation to calculate the amount of the other substance.

Step 3: Convert this amount to a mass or a volume, depending on what the question asks.

EXAM HINT

When attempting multistep calculations involving different units, it is important to show the units for each step of the reaction. It will force you to think about what you have just calculated rather than leaving a number floating.

EXAMPLE 4

A piece of magnesium with a mass of 1.00 g is added to an excess of dilute hydrochloric acid. What volume of hydrogen gas is formed?

The equation for the reaction is:

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

You are not given any information about the hydrochloric acid, and you are not asked anything about magnesium chloride. You can use the mole expression to calculate the amount of magnesium:

$$n(\text{Mg}) = \frac{1.00}{24.3} = 0.0412\,\text{mol}$$

You can see that the Mg : H_2 ratio in the equation is 1 : 1, which means that 0.0412 mol of hydrogen is formed.

Convert this amount to a volume and you have the answer:

volume = $24 \times 0.0412 = 0.99\,dm^3$

EXAMPLE 5

Calcium carbonate reacts with nitric acid to form calcium nitrate, water and carbon dioxide, as shown in the equation:

$$CaCO_3(s) + 2HNO_3(aq) \rightarrow Ca(NO_3)_2(aq) + H_2O(l) + CO_2(g)$$

In a reaction, 100 cm^3 of carbon dioxide is formed. What mass of calcium carbonate is needed for this?

You are not told anything about nitric acid, or asked anything about calcium nitrate or water. You can use the molar volume expression to calculate the amount of carbon dioxide:

$$\text{amount} = \frac{100}{24000} = 0.00417\,\text{mol}$$

You can see that the $CaCO_3$: CO_2 ratio in the equation is 1 : 1, which means that 0.00417 mol of calcium carbonate is needed.

Convert this amount to a mass and you have the answer:

$m = n \times M = 0.00417 \times 100.1 = 0.42\,g$

EXAMPLE 6

Ammonium sulfate reacts with sodium hydroxide solution to form sodium sulfate, water and ammonia, as shown in the equation:

$$(NH_4)_2SO_4(s) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l) + 2NH_3(g)$$

What volume of ammonia is formed by reacting 2.16 g of ammonium sulfate with excess sodium hydroxide solution?

You are not given any information about the sodium hydroxide, and you are not asked anything about sodium sulfate or water. You can use the mole expression to calculate the amount of ammonium sulfate:

$$n((NH_4)_2SO_4) = \frac{2.16}{132.1} = 0.01635 \text{ mol}$$

You can see that the $(NH_4)_2SO_4 : NH_3$ ratio in the equation is 1 : 2, which means that $0.01635 \times 2 = 0.0327$ mol of ammonia is formed.

Convert this amount to a volume and you have the answer:

$$\text{volume} = 24\,000 \times 0.0327 = 785 \text{ cm}^3$$

LEARNING TIP

Practise using the three-step method for calculating masses from volumes, and volumes from masses, in some reactions.

CHECKPOINT

SKILLS PROBLEM SOLVING

In these questions, assume that all volumes are measured at r.t.p.

1. A flask contains 2 dm^3 of butane. What is the amount, in moles, of gas in the flask?

2. 10.0 g of copper(II) oxide is heated with hydrogen according to this equation:

$$CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(l)$$

What volume of hydrogen gas is needed to react with the copper(II) oxide, and what mass of copper is formed?

SUBJECT VOCABULARY

molar volume the volume occupied by 1 mol of any gas; this is normally 24 dm^3 or 24 000 cm^3 at r.t.p.

1E 2 CONCENTRATIONS OF SOLUTIONS

SPECIFICATION REFERENCE 1.5

LEARNING OBJECTIVES

- Calculate the concentration of a solution, in g dm^{-3} and mol dm^{-3}.

CALCULATIONS USING MASS CONCENTRATION (g dm^{-3})

If you know the mass of a **solute** that you dissolve in a **solvent** (usually water), and the volume of the **solution** formed, then it is straightforward to calculate the **mass concentration**.

You use the expression:

$$\text{mass concentration in g dm}^{-3} = \frac{\text{mass of solute in g}}{\text{volume of solution in dm}^3}$$

In this topic, we only use values based on g and dm^3. You may sometimes see other units, such as g cm^{-3} and kg m^{-3}.

As with other similar expressions, you will need to rearrange it, depending on the wording of the question. You also need to remember to convert cm^3 to dm^3 (by dividing by 1000).

EXAMPLE 1

200 cm^3 of a solution contains 5.68 g of sodium bromide. What is its mass concentration?

$$\text{mass concentration} = \frac{m}{V} = \frac{5.68}{0.200} = 28.4\,\text{g dm}^{-3}$$

EXAMPLE 2

The concentration of a solution is 15.7 g dm^{-3}. What mass of solute is there in 750 cm^3 of solution?

$$m = \text{mass concentration} \times V = 15.7 \times 0.750 = 11.8\,\text{g}$$

EXAMPLE 3

A chemist uses 280 g of a solute to make a solution of concentration 28.4 g dm^{-3}. What volume of solution does he make?

$$V = \frac{m}{\text{mass concentration}} = \frac{280}{28.4} = 9.86\,\text{dm}^3$$



▲ **fig A** These containers indicate increasing relative solute concentrations by increasing colour intensity. Unfortunately, most solutions we use are not coloured, so we cannot rely on different colour intensities to indicate different concentrations.

CALCULATIONS USING MOLAR CONCENTRATION (mol dm^{-3})

Molar concentration (it used to be called molarity) is used more often than mass concentration. If only the term 'concentration' is mentioned, then you should assume that it refers to molar concentration.

The units of molar concentration are mol dm^{-3}, and this is often denoted by using square brackets. If a solution of hydrochloric acid has a concentration of 0.150 mol dm^{-3}, this can be shown as [HCl] = 0.150 mol dm^{-3}. The symbol c is sometimes used to represent molar concentration.

You need to be able to use these two expressions together:

$$\text{amount} = \frac{\text{mass}}{\text{molar mass}} \text{ or } n = \frac{m}{M}$$

and

$$\text{(molar) concentration} = \frac{\text{amount}}{\text{volume}} \text{ or } c = \frac{n}{V}$$

As previously, you may need to rearrange these expressions. Look at the question wording to decide which one to use first.

WORKED EXAMPLE 1

A chemist makes 500 cm^3 of a solution of nitric acid of concentration 0.800 mol dm^{-3}. What mass of HNO_3 does she need?

Step 1: You are given values of V and c, so you can use the second expression to calculate a value for n.

$$n = c \times V = 0.800 \times 0.500 = 0.400\,\text{mol}$$

Step 2: You can now use the first expression to calculate the mass of nitric acid.

$$m = n \times M = 0.400 \times 63.0 = 25.2\,\text{g}$$

WORKED EXAMPLE 2

A student has 50.0 g of sodium chloride. What volume of a 0.450 mol dm^{-3} solution can he make?

Step 1: You are given the value of m and can work out M from the Periodic Table, so you can calculate n.

$$n = \frac{m}{M} = \frac{50.0}{58.5} = 0.855\,\text{mol}$$

Step 2: You can now use the second expression to calculate the volume of solution.

$$V = \frac{n}{c} = \frac{0.855}{0.450} = 1.90\,\text{dm}^3$$

CALCULATIONS FROM EQUATIONS USING CONCENTRATION AND MASS

In this type of calculation, you can use an equation to calculate the mass of a reactant or product if you are given the volume and molar concentration of another substance, and vice versa.

The expressions you need are the same as those you have just used, but you also need the equation for the reaction so that you can see the reacting ratio.

WORKED EXAMPLE 3

An excess of magnesium is added to 100 cm^3 of 1.50 mol dm^{-3} hydrochloric acid. The equation for the reaction is:

$$Mg + 2HCl \rightarrow MgCl_2 + H_2$$

What mass of hydrogen is formed?

Step 1: You are given the values of V and c, so you can use the second expression to calculate the value of n for hydrochloric acid.

$$n = 0.100 \times 1.50 = 0.150 \text{ mol}$$

Step 2: The ratio for $HCl : H_2$ is 2 : 1, so $n(H_2) = 0.150 \div 2 = 0.0750$ mol

Step 3: For hydrogen, $m = n \times M = 0.0750 \times 2.0 = 0.15$ g

WORKED EXAMPLE 4

A mass of 47.8 g of magnesium carbonate reacts with 2.50 mol dm^{-3} hydrochloric acid. The equation for the reaction is:

$$MgCO_3 + 2HCl \rightarrow MgCl_2 + H_2O + CO_2$$

What volume of acid is needed?

Step 1: You are given the value of m and can work out M from the Periodic Table, so you can calculate n.

$$n = \frac{47.8}{84.3} = 0.567 \text{ mol}$$

Step 2: The ratio for $MgCO_3 : HCl$ is 1 : 2, so $n(HCl) = 2 \times 0.567 = 1.134$ mol

Step 3: for HCl, $V = \frac{1.134}{2.50} = 0.454 \text{ dm}^3$

LEARNING TIP

Remember that the volumes referred to in this topic are of solutions, not solvents. If you dissolve a solute in 100 cm^3 of a solvent, the volume of the solution is not exactly 100 cm^3.

CHECKPOINT

SKILLS PROBLEM SOLVING

1. 50.0 g of sodium hydroxide is dissolved in water to make 1.50 dm^3 of solution. What is the molar concentration of the solution?

2. 150 cm^3 of 0.125 mol dm^{-3} lead(II) nitrate solution is mixed with an excess of potassium iodide solution. The equation for the reaction that occurs is:

$$Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$$

What mass of lead(II) iodide is formed?

SUBJECT VOCABULARY

solute a substance that is dissolved
solvent a substance that dissolves a solute
solution a solute dissolved in a solution
mass concentration (of a solution) the mass (in g) of the solute divided by the volume of the solution
molar concentration (of a solution) the amount (in mol) of the solute divided by the volume of the solution

1E 3 CONCENTRATIONS IN PPM

SPECIFICATION REFERENCE 1.4(iv)

LEARNING OBJECTIVES

- Understand the term parts per million (ppm), e.g. gases in the atmosphere.

CONCENTRATIONS OF SOLUTIONS

Concentrations of solutions can also be compared using the term **parts per million**, or **ppm**. It is often used for pollutants in water.

Think of the word 'percent' – the 'cent' part refers to 100, so 50% means 50 out of 100, and parts per million means out of a million. 50 ppm means 50 parts out of 1 000 000 parts. The 'parts' usually refer to mass. The parts can be measured in any unit of mass, but grams or milligrams are the most commonly used.

CALCULATIONS FOR SOLUTIONS IN PPM

A concentration of 1 ppm means 1 g in 1 000 000 g, or 1 mg in 1 000 000 mg.

This expression can be used to calculate concentrations in ppm:

$$\text{concentration in ppm} = \frac{\text{mass of solute} \times 1\,000\,000}{\text{mass of solvent}}$$

The masses can be in any units, but they must be the same units. If different units are given, then one of them must be converted.

WORKED EXAMPLE 1

A solution contains 0.176 g of solute dissolved in 750 g of solvent. What is the concentration in ppm?

As the units of solute and solvent are the same, the values can be directly inserted into the expression.

$$\text{concentration in ppm} = \frac{\text{mass of solute} \times 1\,000\,000}{\text{mass of solvent}}$$

$$= \frac{0.176 \times 1\,000\,000}{750} = 235 \text{ ppm}$$

WORKED EXAMPLE 2

A mass of 23 mg of sodium chloride is dissolved in 900 g of water. What is the concentration of sodium chloride in the solution in ppm?

As the units of solute and solvent are different, either the mass of solute or the mass of solvent must be converted so they are the same. Then the values can be directly inserted into the expression.

First, convert the mass of sodium chloride from mg to g (divide by 1000):

mass of sodium chloride = 23 ÷ 1000 = 0.023 g

$$\text{concentration in ppm} = \frac{\text{mass of solute} \times 1\,000\,000}{\text{mass of solvent}}$$

$$= \frac{0.023 \times 1\,000\,000}{900} = 26 \text{ ppm}$$

WORKED EXAMPLE 3

A sample of river water contains phosphate ions with a concentration of 17 ppm. What is the mass of phosphate ions in 500 g of the river water?

In this example, the expression needs to be rearranged:

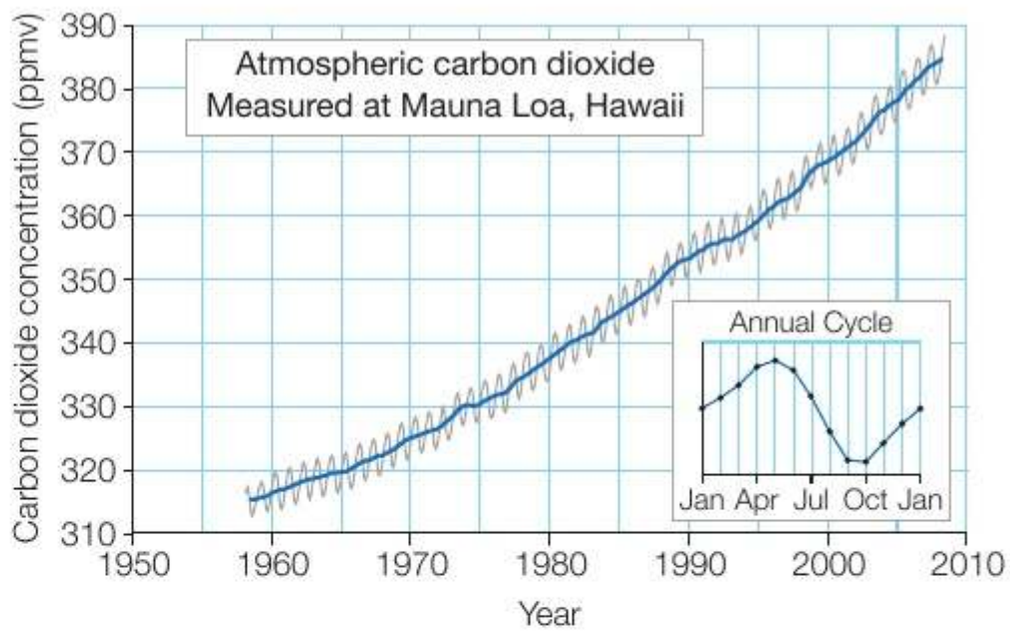
$$\text{mass of solute} = \frac{\text{concentration in ppm} \times \text{mass of solvent}}{1\,000\,000}$$

$$= \frac{17 \times 500}{1\,000\,000} = 0.0085 \text{ g}$$

GASES IN THE ATMOSPHERE

Mauna Loa is the name of a volcano in Hawaii. It is also the location of a weather observatory that has recorded the levels of carbon dioxide in the atmosphere over a long period of time.

Fig A shows how the levels of carbon dioxide at Mauna Loa have changed over a period of 50 years. There are small variations during each year. In 1960, the level was about 316 ppm. By 2010, the level had reached nearly 390 ppm. Since 2015 it has been over 400 ppm, and many scientists believe that it will always remain above this value.



▲ **fig A** This graph shows the steady increase in atmospheric carbon dioxide over five decades.

The concentrations of pollutant gases in the atmosphere are often given as values in ppm. Instead of using masses, the comparison is usually by volume, so calculations are done in a different way. Sometimes the values are quoted in ppmv – the v shows that the value refers to concentration by volume.

A concentration of 1 ppmv means 1 cm^3 in 1 000 000 cm^3, or 1 cm^3 in 1 000 dm^3.

Residents of Dhaka, the capital city of Bangladesh, are concerned about air quality. As in many cities, road traffic leads to high levels of carbon monoxide, especially during rush hours, when many people are travelling to and from work. Levels of CO in the air can be as high as 100 ppm. Outside the city there are many factories producing bricks for constructing buildings. These factories are responsible for many pollutants, including sulfur dioxide and particulate matter.



▲ **fig B** This brick factory releases pollutant gases into the atmosphere.

CALCULATIONS FOR GASES IN PPM

This expression can be used to calculate concentrations in ppm.

$$\text{concentration in ppm} = \frac{\text{volume of gas} \times 1\,000\,000}{\text{volume of air}}$$

The volumes can be in any units, but they must be the same units. If different units are given, then one of them must be converted.

WORKED EXAMPLE 4

Some nitrogen dioxide gas, with a volume of 1.5 dm^3, mixes with 10 000 dm^3 of air. What is the concentration of nitrogen dioxide, in ppm, in the air?

As the volume units of both gases are the same, then the values can be directly inserted into the expression.

$$\text{concentration in ppm} = \frac{\text{volume of gas} \times 1\,000\,000}{\text{volume of air}}$$

$$= \frac{1.5 \times 1\,000\,000}{10\,000} = 150 \text{ ppm}$$

WORKED EXAMPLE 5

5000 dm^3 of air is found to contain ozone with a concentration of 87 ppm. What volume of ozone is in this sample of air?

In this example, the expression needs to be rearranged:

$$\text{volume of gas} = \frac{\text{concentration in ppm} \times \text{volume of air}}{1\,000\,000}$$

$$= \frac{87 \times 5000}{1\,000\,000} = 0.435 \text{ dm}^3$$

WORKED EXAMPLE 6

Two samples of air containing sulfur dioxide were analysed. The results for Sample 1 showed that 500 dm^3 of air contained 37 cm^3 of sulfur dioxide. The results for Sample 2 showed that there were 1.4 dm^3 of sulfur dioxide in 4000 dm^3 of air. Show, by calculation, which sample has the higher concentration, in ppm, of sulfur dioxide.

Sample 1

$$\text{concentration in ppm} = \frac{\text{volume of gas} \times 1\,000\,000}{\text{volume of air}}$$

$$= \frac{37 \div 1000 \times 1\,000\,000}{500} = 74 \text{ ppm}$$

Sample 2

$$\text{concentration in ppm} = \frac{\text{volume of gas} \times 1\,000\,000}{\text{volume of air}}$$

$$= \frac{1.4 \times 1\,000\,000}{4000} = 350 \text{ ppm}$$

Sample 2 has the higher concentration.

LEARNING TIP

For ppm calculations with solutions, use the expression involving masses. For calculations with gases, use the expression involving volumes.

CHECKPOINT

1. 0.2 g of potassium sulfate is dissolved in water to make 800 g of solution. What is the concentration of the salt in ppm?
2. 200 dm^3 of air contains 58 cm^3 of chlorine. What is the concentration of chlorine in ppm?

SUBJECT VOCABULARY

parts per million (ppm) the number of parts of one substance in one million parts of another substance; a measure used to describe chemical concentration; usually, 'parts' refers to masses of both substances, or to volumes of both substances